

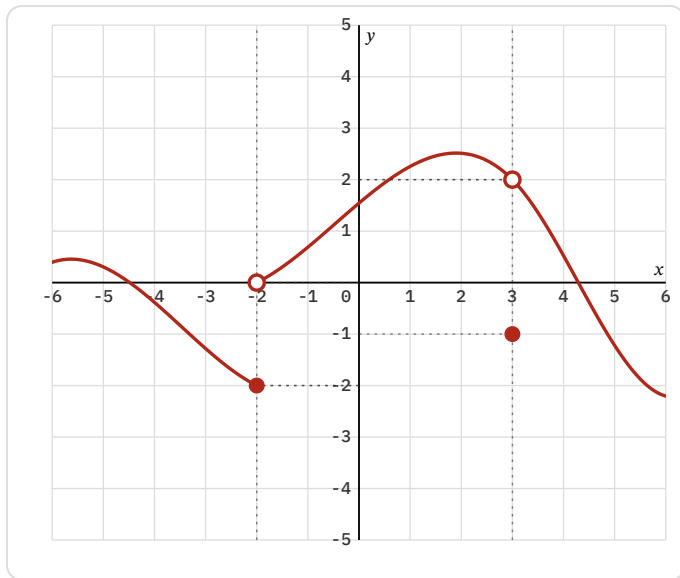
Mock Midterm B — Answer Key

Practice midterm (AI-Free) · Sections 1.1–1.4

Total: 96 points

1.

Use the graph of $y = f(x)$ to state the value of each quantity, if it exists. If it does not exist, write **DNE**.



a.) $f(-2)$ (1)

-2

b.) $\lim_{x \rightarrow -2^-} f(x)$ (1)

-2

c.) $\lim_{x \rightarrow -2^+} f(x)$ (1)

0

d.) $\lim_{x \rightarrow -2} f(x)$ (1)

DNE

e.) $f(3)$ (1)

-1

f.) $\lim_{x \rightarrow 3^-} f(x)$ (1)

2

g.) $\lim_{x \rightarrow 3^+} f(x)$ (1)

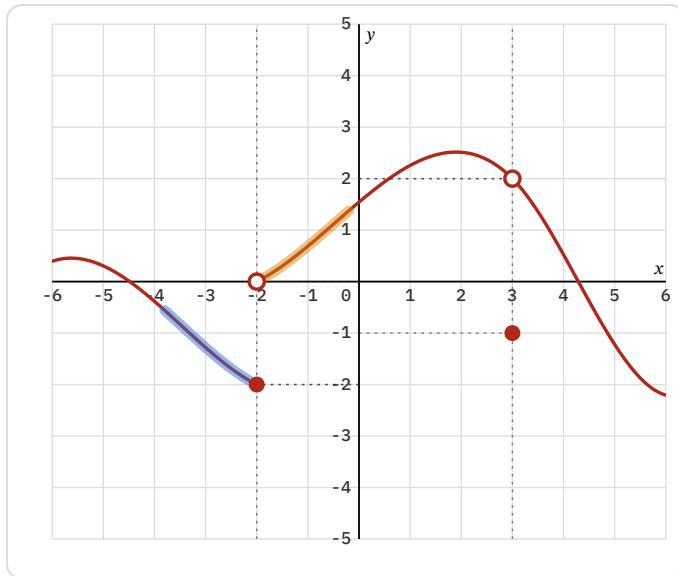
2

h.) $\lim_{x \rightarrow 3} f(x)$ (1)

2

8 pts

WORKED SOLUTION



— approaching $x = -2$ from the left — from the right. Follow each colored piece of the curve to the height it heads toward.

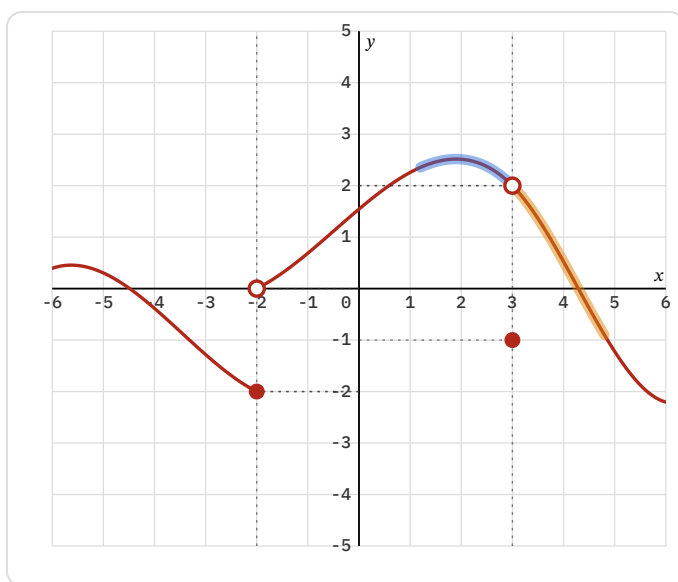
At $x = -2$

The **filled** dot at $x = -2$ sits at height -2, so $f(-2) = -2$.

Tracing the curve toward $x = -2$ from the left, the heights approach -2 (open or closed dot doesn't matter).

Tracing the curve toward $x = -2$ from the right, the heights approach 0 (open or closed dot doesn't matter).

The left and right limits don't match, so $\lim_{x \rightarrow -2} f(x)$ DNE.



— approaching $x = 3$ from the left — from the right. Follow each colored piece of the curve to the height it heads toward.

At $x = 3$

The **filled** dot at $x = 3$ sits at height -1, so $f(3) = -1$.

Tracing the curve toward $x = 3$ from the left, the heights approach 2 (open or closed dot doesn't matter).

Tracing the curve toward $x = 3$ from the right, the heights approach 2 (open or closed dot doesn't matter).

Both one-sided limits equal 2, so $\lim_{x \rightarrow 3} f(x) = 2$ — even though $f(3) = -1$. The limit ignores the value AT the point.

2. Let $f(x) = \frac{e^{-x}-1}{x}$. Use the table to see what the outputs do as x approaches 0.

7 pts

x	-0.5	-0.1	-0.01	0	0.01	0.1	0.5
$f(x)$	-1.297443	-1.051709	-1.005017	?	-0.995017	-0.951626	-0.786939

a.) Can you find $\lim_{x \rightarrow 0} f(x)$ just by plugging in $x = 0$? (1)

(B) No — $x = 0$ makes a denominator 0, so $f(0)$ is undefined

b.) As $x \rightarrow 0^-$, $f(x)$ approaches... (2)

-1

c.) As $x \rightarrow 0^+$, $f(x)$ approaches... (2)

-1

d.) $\lim_{x \rightarrow 0} f(x) =$ (2)

-1

WORKED SOLUTION

Reading the left three columns, the outputs head toward -1; the right three head toward -1.

Both sides agree, so the limit is -1.

$$\lim_{x \rightarrow 0} \frac{e^{-x} - 1}{x} = -1$$

Why: **no algebra trick here: the table values settle at -1**

3.

Calculate each limit algebraically.

6 pts

$$\text{a.) } \lim_{x \rightarrow 3} (-3x^2 + 4x + 5) \quad (2)$$
$$-10$$

$$\text{b.) } \lim_{x \rightarrow 1} \frac{x^2 - 2x - 1}{x - 4} \quad (2)$$
$$\frac{2}{3}$$

$$\text{c.) } \lim_{x \rightarrow -4} \sqrt{x^2 + 33} \quad (2)$$
$$7$$

WORKED SOLUTION

Each input is in the domain of its function, so direct substitution works.

$$\text{a. } \lim_{x \rightarrow 3} (-3x^2 + 4x + 5) = -3(3)^2 + 4(3) + 5 = -10$$

$$\text{b. } \lim_{x \rightarrow 1} \frac{x^2 - 2x - 1}{x - 4} = \frac{(1)^2 - 2(1) - 1}{1 - 4} = \frac{-2}{-3} = \frac{2}{3}$$

$$\text{c. } \lim_{x \rightarrow -4} \sqrt{x^2 + 33} = \sqrt{(-4)^2 + 33} = \sqrt{49} = 7$$

4.

Calculate each limit algebraically. (Direct substitution gives $\frac{0}{0}$ — simplify first.) If it doesn't exist, write DNE.

a.) $\lim_{x \rightarrow -1} \frac{x+1}{x^2+2x+1}$ (2)
DNE

b.) $\lim_{h \rightarrow 0} \frac{\frac{1}{3+h} - \frac{1}{3}}{h}$ (2)
 $-\frac{1}{9}$

4 pts

WORKED SOLUTION

a.

$$\lim_{x \rightarrow -1} \frac{x+1}{x^2+2x+1} = \lim_{x \rightarrow -1} \frac{(x+1)}{(x+1)^2} = \lim_{x \rightarrow -1} \frac{1}{x+1}. \text{ Left } \rightarrow -\infty, \text{ right } \rightarrow \infty, \text{ so the limit DNE}$$

b.

$$\lim_{h \rightarrow 0} \frac{\frac{1}{3+h} - \frac{1}{3}}{h} = \lim_{h \rightarrow 0} \frac{\frac{3-(3+h)}{3(3+h)}}{h} = \lim_{h \rightarrow 0} \frac{-h}{3(3+h)h} = \lim_{h \rightarrow 0} \frac{-1}{3(3+h)} = -\frac{1}{9}$$

Notation tip: keep writing **lim** on every line until the step where you actually substitute.

5. Compute each limit algebraically. Use $\infty / -\infty$ for unbounded results.

6 pts

a.) $\lim_{x \rightarrow \infty} \frac{-4x^3 + 1}{4x^4 - 9x^2 - 9x - 1}$ (2)
 0

b.) $\lim_{x \rightarrow -\infty} \frac{-2x^4 + 9x - 5}{x^4 + 7x^3 + 7x - 3}$ (2)
 -2

c.) $\lim_{x \rightarrow \infty} \frac{7x^5 - 2x^4 - 5}{3x^3 - 4x^2 - 6x + 5}$ (2)
 ∞

WORKED SOLUTION

a. Divide every term by x^4 , the highest power of x in the denominator:

$$\lim_{x \rightarrow \infty} \frac{-4x^3 + 1}{4x^4 - 9x^2 - 9x - 1} = \lim_{x \rightarrow \infty} \frac{-\frac{4}{x} + \frac{1}{x^4}}{4 - \frac{9}{x^2} - \frac{9}{x^3} - \frac{1}{x^4}}$$

The top $\rightarrow 0$ while the bottom $\rightarrow 4$, so the limit is 0 .

b. Divide every term by x^4 , the highest power of x in the denominator:

$$\lim_{x \rightarrow -\infty} \frac{-2x^4 + 9x - 5}{x^4 + 7x^3 + 7x - 3} = \lim_{x \rightarrow -\infty} \frac{-2 + \frac{9}{x^3} - \frac{5}{x^4}}{1 + \frac{7}{x} + \frac{7}{x^3} - \frac{3}{x^4}}$$

Every fraction term $\rightarrow 0$, leaving $\frac{-2}{1} = -2$.

c. Divide every term by x^3 , the highest power of x in the denominator:

$$\lim_{x \rightarrow \infty} \frac{7x^5 - 2x^4 - 5}{3x^3 - 4x^2 - 6x + 5} = \lim_{x \rightarrow \infty} \frac{7x^2 - 2x - \frac{5}{x^3}}{3 - \frac{4}{x} - \frac{6}{x^2} + \frac{5}{x^3}}$$

The bottom $\rightarrow 3$, but the top is dominated by $7x^2$, which $\rightarrow \infty$ as $x \rightarrow \infty$; dividing by 3 gives ∞ .

6.

Let

7 pts

$$G(x) = \begin{cases} x^2 - 3 & x < 0 \\ -2x & x > 0 \end{cases}$$

Find each quantity. If it does not exist, write DNE.

a.) $G(0)$ (1)

DNE

b.) $\lim_{x \rightarrow 0^-} G(x)$ (1)

-3

c.) $\lim_{x \rightarrow 0^+} G(x)$ (1)

0

d.) $\lim_{x \rightarrow 0} G(x)$ (1)

DNE

e.) $G(2)$ (1)

-4

f.) $\lim_{x \rightarrow -\infty} G(x)$ (1)

 ∞

g.) $\lim_{x \rightarrow \infty} G(x)$ (1)

 $-\infty$ **WORKED SOLUTION** **$G(0)$:** the pieces cover $x < 0$ and $x > 0$ only, so $G(0)$ is undefined $\rightarrow$ DNE.**Left:** use $x^2 - 3$: $(0)^2 - 3 = -3$.**Right:** use $-2x$: $-2(0) + 0 = 0$.**Two-sided:** $-3 \neq 0$, so the limit DNE. **$G(2)$:** $2 > 0$, so use $-2x$: $-2(2) + 0 = -4$.**End behavior:** as $x \rightarrow -\infty$ we're on the $x^2 - 3$ piece — an even-degree term with positive leading coefficient $\rightarrow \infty$. As $x \rightarrow \infty$ we're on the line with slope -2 $\rightarrow -\infty$.

7. Let

6 pts

$$f(x) = \begin{cases} \frac{2x^2 - 3x + 1}{2x - 1} & x < \frac{1}{2} \\ 2x - 2 & x > \frac{1}{2} \end{cases}$$

Find each limit or function value algebraically. Write DNE if it doesn't exist.

a.) $\lim_{x \rightarrow 3} f(x)$ (1)

4

b.) $\lim_{x \rightarrow -3} f(x)$ (1)

-4

c.) $f\left(\frac{1}{2}\right)$ (1)

DNE

d.) $\lim_{x \rightarrow \frac{1}{2}^-} f(x)$ (1)

$-\frac{1}{2}$

e.) $\lim_{x \rightarrow \frac{1}{2}^+} f(x)$ (1)

-1

f.) $\lim_{x \rightarrow \frac{1}{2}} f(x)$ (1)

DNE

WORKED SOLUTION

For $x < \frac{1}{2}$, factor the top: $\frac{(2x-1)(x-1)}{2x-1} = x - 1$ (valid since $x \neq \frac{1}{2}$ there).

a. $3 > \frac{1}{2} \rightarrow$ bottom piece: $2(3) - 2 = 4$.

b. $-3 < \frac{1}{2} \rightarrow$ top piece, simplified: $-3 - 1 = -4$.

c. Neither piece includes $x = \frac{1}{2}$, so $f\left(\frac{1}{2}\right)$ is undefined $\rightarrow$ DNE.

d. Left: $\lim_{x \rightarrow \frac{1}{2}^-} (x - 1) = -\frac{1}{2}$.

e. Right: $\lim_{x \rightarrow \frac{1}{2}^+} (2x - 2) = -1$.

f. $-\frac{1}{2} \neq -1$, so the limit DNE.

8.

Find the value of k that makes f continuous at $x = 1$.

4 pts

$$f(x) = \begin{cases} x^2 & x < 1 \\ kx + 5 & x \geq 1 \end{cases}$$

a.) $\lim_{x \rightarrow 1^-} f(x)$ (2)

$$1$$

b.) $k =$ (2)

$$-4$$

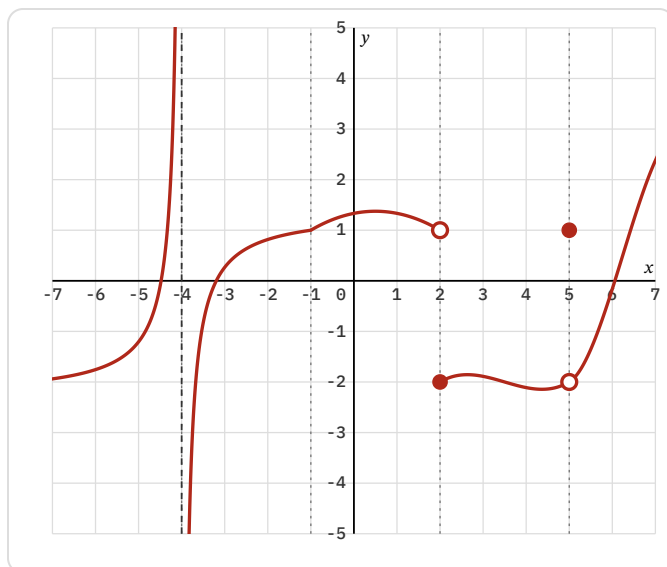
WORKED SOLUTIONLeft: $(1)^2 + 0(1) = 1$. Right (and $f(1)$): $k(1) + 5$.

$$1k + 5 = 1 \Rightarrow k = \frac{-4}{1} = -4$$

9.

Classify the behavior of f at each labeled x -value.

4 pts

a.) At $x = -4$ (1)

(D) Infinite discontinuity

b.) At $x = -1$ (1)

(A) Continuous

c.) At $x = 2$ (1)

(C) Jump discontinuity

d.) At $x = 5$ (1)

(B) Removable discontinuity

WORKED SOLUTION $x = -4$: The graph shoots off along a vertical asymptote $\rightarrow$ infinite. $x = -1$: No break, and the dot is on the curve $\rightarrow$ continuous. $x = 2$: Left approaches 1, right approaches -2 $\rightarrow$ jump. $x = 5$: Both sides approach -2, but the filled dot is at 1 $\rightarrow$ removable (the limit exists, it just doesn't match f).

10.

Let $P(t) = 2t^2 + 4t + 56$, where $P(t)$ represents the population of a town, in thousands of people, t years after 2020. Find the average rate of change over $[1, 4]$.

a.) $P(1)$ (2)

62

b.) $P(4)$ (2)

104

c.) Average rate of change (number only) (2)

14

d.) Which sentence correctly states the result **with units**? (1)

(B) Between $t = 1$ and $t = 4$ years, the population changes at an average rate of 14 thousand people per year.

7 pts

WORKED SOLUTION

$$P(1) = 62, \quad P(4) = 104$$

$$\frac{P(4) - P(1)}{4 - 1} = \frac{104 - (62)}{3} = \frac{42}{3} = 14 \text{ thousand people per year}$$

It's an *average* over an interval (not an instant), and the units are thousand people per year — thousand people per year. The total change would be 42 thousand people.

11.

Let $f(x) = \frac{1}{x}$. Calculate the average rate of change of f over each interval. (Rounded answers to 2+ decimal places are fine.)

a.) $[1, 2]$ (1)

$$-\frac{1}{2}$$

b.) $[1, 1.5]$ (1)

$$-\frac{2}{3}$$

c.) $[1, 1.1]$ (1)

$$-\frac{10}{11}$$

d.) $[1, 1.001]$ (1)

$$-\frac{1000}{1001}$$

e.) As the interval shrinks toward $x = 1$, these values approach... (1)

$$(B) -1$$

5 pts

WORKED SOLUTION

$$[1, 2] : \frac{f(2) - f(1)}{1} = -\frac{1}{2} \approx -0.5$$

$$[1, 1.5] : \frac{f(1.5) - f(1)}{0.5} = -\frac{2}{3} \approx -0.6667$$

$$[1, 1.1] : \frac{f(1.1) - f(1)}{0.1} = -\frac{10}{11} \approx -0.9091$$

$$[1, 1.001] : \frac{f(1.001) - f(1)}{0.001} = -\frac{1000}{1001} \approx -0.999$$

Each value is the slope of a secant line through $(1, f(1))$. As the interval shrinks they approach the slope of the **tangent line**: $f'(x) = -\frac{1}{x^2}$, so $f'(1) = -1$.

12.

Let $f(x) = x^2 + 2x + 5$. Find and simplify the difference quotient $\frac{f(x+h)-f(x)}{h}$, then use it to fill in the table.

a.) Simplified difference quotient (in terms of x and h) (3)

$$2x + h + 2$$

b.) $x = 0, h = 1$ (1)

$$3$$

c.) $x = 0, h = 0.1$ (1)

$$\frac{21}{10}$$

d.) $x = 0, h = 0.01$ (1)

$$\frac{201}{100}$$

e.) As $h \rightarrow 0$, what value do these approach? (1)

$$2$$

7 pts

WORKED SOLUTION

$$f(x+h) = (x+h)^2 + 2(x+h) + 5 = x^2 + 2xh + h^2 + 2x + 2h + 5$$

$$f(x+h) - f(x) = 2xh + h^2 + 2h$$

$$\frac{f(x+h) - f(x)}{h} = \frac{h(2x + h + 2)}{h} = 2x + h + 2$$

Now substitute:

$$x = 0, h = 1 : 3$$

$$x = 0, h = 0.1 : \frac{21}{10} = 2.1$$

$$x = 0, h = 0.01 : \frac{201}{100} = 2.01$$

Letting $h \rightarrow 0$ leaves **2**: the instantaneous rate of change (slope of the tangent line) at $x = 0$. That's exactly what section 1.4 formalizes.

13.

Let $f(x) = \frac{-4}{x}$. Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$.

a.) $f'(x) =$ (3)
 $\frac{4}{x^2}$

b.) $f'(2) =$ (2)
 1

c.) Slope of the tangent line to f at $x = -1$ (2)
 4

d.) What does $f'(2) = 1$ mean? (1)

(D) The instantaneous rate of change of f at $x = 2$ is 1 — the slope of the tangent line there.

8 pts

WORKED SOLUTION

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{\frac{-4}{x+h} - \frac{-4}{x}}{h} = \lim_{h \rightarrow 0} \frac{\frac{-4x - (-4)(x+h)}{x(x+h)}}{h} \\ &= \lim_{h \rightarrow 0} \frac{4h}{x(x+h)} \cdot \frac{1}{h} = \lim_{h \rightarrow 0} \frac{4}{x(x+h)} = \frac{4}{x^2} \end{aligned}$$

Then $f'(2) = 1$ and the slope of the tangent at $x = -1$ is $f'(-1) = 4$.

$f'(2)$ is the *instantaneous* rate of change at one input — the slope of the tangent line, not the height of the graph and not an average over an interval.

14. Let $f(x) = x^3 + 2x$. Find the equation of the tangent line to the graph of f at $x = 1$. 7 pts

a.) $f(1)$ (2)
 3

b.) $f'(1)$ (2)
 5

c.) Tangent line (3)
 $y = 5x - 2$

WORKED SOLUTION

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{[(x+h)^3 + 2(x+h)] - (x^3 + 2x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{3x^2h + 3xh^2 + h^3 + 2h}{h} = \lim_{h \rightarrow 0} (3x^2 + 3xh + h^2 + 2) = 3x^2 + 2 \end{aligned}$$

Point: $f(1) = 3$, so the point is $(1, 3)$. Slope: $m = f'(1) = 5$.

$$y - 3 = 5(x - 1) \Rightarrow y = 5x - 2$$

Answers in point-slope form are accepted too.

15.

Let $h(t) = -16t^2 + 160t$ be the height, in feet, of a rocket t seconds after launch.

a.) $h'(t) =$ (3)
 $-32t + 160$

b.) $h'(1)$ (2)
 128

c.) Interpret $h'(1) = 128$. (1)

(C) At $t = 1$ seconds, the height of the rocket is increasing at a rate of 128 feet per second.

6 pts

WORKED SOLUTION

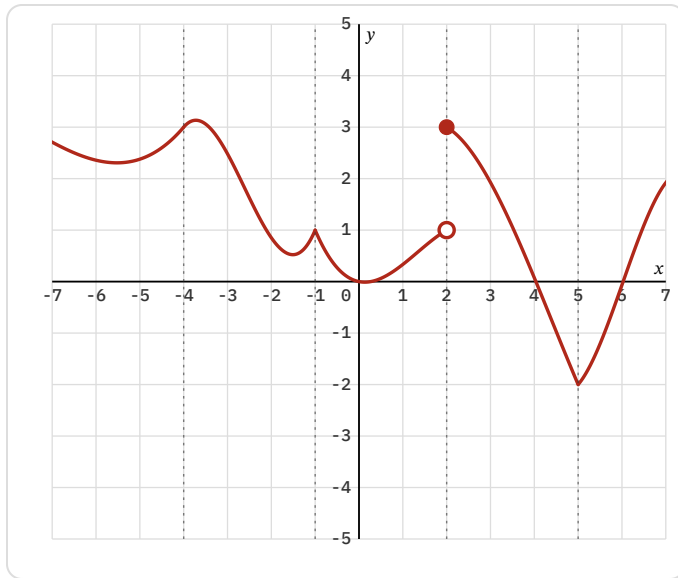
$$\begin{aligned} h'(t) &= \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h} = \lim_{h \rightarrow 0} \frac{[-16(t+h)^2 + 160(t+h)] - (-16t^2 + 160t)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-16t^2 - 32th - 16h^2 + 160t + 160h + 16t^2 - 160t}{h} \\ &= \lim_{h \rightarrow 0} \frac{-32th - 16h^2 + 160h}{h} = \lim_{h \rightarrow 0} \frac{h(-32t - 16h + 160)}{h} \\ &= \lim_{h \rightarrow 0} (-32t - 16h + 160) = -32t + 160 \end{aligned}$$

$$h'(1) = -32(1) + 160 = 128$$

Positive/negative tells the direction, and the units are feet per second: at that instant the height of the rocket is increasing at 128 feet per second. (The value $h(1) = 144$ is a different quantity.)

16.

At each labeled x -value, decide whether the graph of f is continuous there and whether it is differentiable there.



- a.) At $x = -4$ (1)
(A) Continuous and differentiable
- b.) At $x = -1$ (1)
(B) Continuous but NOT differentiable
- c.) At $x = 2$ (1)
(C) Not continuous (so not differentiable)
- d.) At $x = 5$ (1)
(B) Continuous but NOT differentiable

4 pts

WORKED SOLUTION

$x = -4$: Smooth and unbroken — a single tangent line fits → continuous and differentiable.

$x = -1$: No break, but the curve comes to a sharp point (corner): the slopes from the left and right disagree → continuous, not differentiable.

$x = 2$: The graph jumps → not continuous, so not differentiable.

$x = 5$: No break, but the curve comes to a sharp point (corner): the slopes from the left and right disagree → continuous, not differentiable.
